

MAE101 - Statics and Strength of Materials

With Solved Examples

Summer 2026

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1 External Forces

When we talk about forces acting on an object and free body diagrams, we assume these are the only forces on the object. This however is not necessarily true. We shall make the following distinction:

1. External forces are those that act upon the object/system from outside
2. Internal forces are those experienced by members of the object

In equilibrium situations, both the internal and external forces balance; however, in free body diagrams, we only display the external forces (we assume the internal forces are in equilibrium even though the external forces may not).

Throughout this series of notes, we shall explicitly distinguish the system from the environment.

1.1 Geometric Vector Addition

While algebraic vector addition is no doubt important, geometric methods help us to visualize the problem, especially when it comes to resolving force components about non-orthogonal axes. Consider the two vectors:

